SPIN RESONANCE SPECTROSCOPY

2018

7. (a) Which of the following nuclei can exhibit NMR spectrum:

^{37}Cl = NMR active.

^{12}C = NMR inactive.

^{1}H = NMR active

^{13}C = NMR active.

^{19}F = NMR active.

^{16}O = NMR inactive

^{31}P = NMR active.

(b) Describe an expression ~~beta~~ to determine the extent of interaction betn spin and magnetic field.

Answer:

Angular momentum $I = \sqrt{I(I+1)}\,\frac{h}{2\pi}$

$= \sqrt{I(I+1)}$ units.

$I = 0, \frac{1}{2}, 1, \frac{3}{2}$ ... etc.

$I_z = I, I-1, \ldots 0, -(I-1), I$ (for I integral)

$I_z = I, (I-1), \ldots \frac{1}{2}, -\frac{1}{2} \ldots I$ (for I half integral)

The size of the dipole (for a point charge) can be shown as

$$\mu = \frac{qv}{2m} \cdot \vec{I} = \frac{q\sqrt{I(I+1)}}{2m} \cdot \frac{h}{2\pi}$$

$$= \frac{qh}{4\pi m} \sqrt{I(I+1)} \; Am^2$$

where, q is the charge and m = mass of the spinning particle.

$$Am^2 = kg\, s^{-2} T^{-1} \times m^2$$
$$= J T^{-1}$$

$$\mu = \frac{qh}{4\pi m} \sqrt{I(I+1)} \; J T^{-1} \quad \text{---} (1)$$

For (electron and nuclei), $\mu = \frac{G q h}{4\pi m} \sqrt{I(I+1)} \; J T^{-1}$ ---(2)

For electron "G" will be replaced by "g" which is known as Lande splitting factor.

$$\mu = -g\beta\sqrt{I(I+1)}$$

where, $\beta = \frac{qh}{4\pi m}$ = Bohar magneton.

electronic charge $= 1.60 \times 10^{-19}$ C, mass $= 9.11 \times 10^{-31}$ kg

and $\beta_{electron} = 9.273 \times 10^{-24} J T^{-1}$.

For nuclei, neelear magneton

$$\beta_N = \frac{eh}{4\pi m_p} = 5.050 \times 10^{-27} J T^{-1}.$$

For a nucleous of Mass=M and charge = Pe.

$$\mu = \frac{g_I (Pe)}{2M} \sqrt{I(I+1)} \cdot \frac{h}{2\pi}$$

$$= g \beta_N \sqrt{I(I+1)} \quad JT^{-1} \cdot N$$

(c) If a particular NMR instrument operates at 300.26 MHz, calculate the magnetic field strength which is required for ^1H nucleous to resonance at this frequency. Given that plank constant h is $= 6.62 \times 10^{-34}$ JS. and $\beta_N = 5.05 \times 10^{-27} JT^{-1}$. $\boxed{g = 5.585}$

Answer:

~~$\mu = g \beta_N I$~~

$$\omega = \frac{g \beta_N B_z}{h}$$

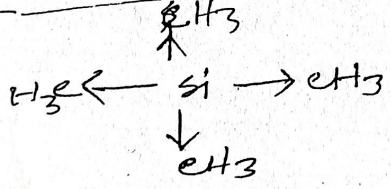
$$\Rightarrow B_z = \frac{\omega h}{g \beta_N} = \frac{300.26 \times 10^6 \times 6.62 \times 10^{-34}}{5.585 \times 5.05 \times 10^{-27}}$$

$$= 7.05 T$$

(d) Why is TMS almost universally used as a standard reference compound for H resonances in NMR spectroscopy.

Answer:

1. Highly volatile liquid. (b.p = 26.5°c)
2. Inert to most organic & reagent / chemical.
3. soluble in most organic solvent.
4. Sharp peak by adding small quantity.
5. The electronegativity of c is more than si and H is more electronegative than si, so H is more shelded iso and 12 H are equivalent so, H given a single peak in upfield region.



H given a peak in upfield region.

(e) What is meant by chemical shift? How would you differentiate trans and cis-isomers with the concept of chemical shift.?

Answer:

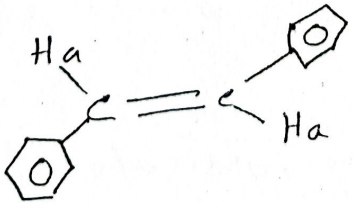
Chemical shift: The value of (δ) is called the chemical shift.

Or, The spe separation betn absorption peaks is usually reffered as ~~absorption peaks~~ chemical shift.

Differentiate of geometric isomer with the concept of chemical shift values of protons.

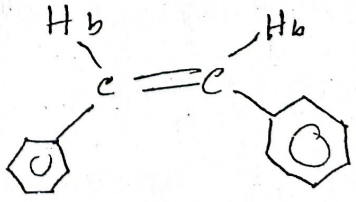
By using the values of chemical shift we differentiate geometric isomer such as cis and trans. The proton that are attached with halogen, C=C, C=O etc gives higher values of δ δ because of these protons placed in desheilded region.

consider a example such as cis-stibene and trans stilbene.



$\delta_{Ha} = 6.99$

trans



$\delta_{Hb} = 6.49$

cis

In trans stilbene the δ_{Ha} is higher and $\delta_{Ha} = 6.09$ than $\delta_{Hb} = 6.90$ in cis stilbene. Because both Ha protons are deshielded by both benzene ring and gives higher δ value. On the other hand, δ_{Hb} value for cis-stilbene gives lower values and deshielded by only one aromatic ring.

(f) What is ESR. What are the criteria for E.S.R spectroscopy? Discuss with examples of molecules / ions / radicals which can exhibit ESR spectrum.

E.S.R : It is a branch of absorption spectroscopy in which radiation of having frequency in microwave region is absorbed by paramagnetic substance to induce transition betn magnetic energy level of electron with unpaired spin.

Absorption spectroscopy, operate at microwave frequency 109-106 MHz.

Criteria:

Some atoms and molecules contain one or more electrons with unpaired spin and these are the substances which are expected to show electron spin resonance spectroscopy.

Example:

(i) simple molecules, NO, O_2 and NO_2

(ii) Ions of transition metal.

Fe^{3+}, $Fe(CN)_6]^{3-}$ etc.

(iii) Unstable paramagnetic materials such as free radicals or radicals.

2017

7. (a) what type of nuclei can exhibit NMR spectrum? Give examples.

Answer:

If the no. of protons and neutrons is odd, then the nucleous ~~is~~ N. exhibit NMR spectrum.

Example: ^{1}H, ^{13}C.

^{1}H = the no. of proton is odd.
the no. of neutron is 0.

^{13}C = the no. of proton + neutron = 13, odd.
so. NMR active.

(b) See 2017.

(c) What is meant by Larmor precessional frequency? show that the Laromor prenional frequency of a spining nucleous in a magnetic field is just equal to the freqency separation betn energy levels.

Answer:

The larmor frequency : When magnetic field is applied, the nucleous begin to precess about its own axis of spin with angular frequency 'ω' which is refferred to as precisional frequency.

Larmor frequency $\omega = \frac{\text{magnetic moment} \times B_z \text{ rads}^{-1}}{\text{angular momentum}}$

$$= \frac{\mu B_z}{2\pi I} \text{ Hz} \quad \text{---} (1)$$

The dipole moment of a spining nucleous—

$$\mu = g \beta_N \sqrt{I(I+1)} \text{ JT}^{-1} \quad \text{---} (2)$$

∴ From eqn (1) we get,

$$\omega = \frac{g \beta_N \sqrt{I(I+1)} \cdot B_z}{\sqrt{I(I+1)} \cdot \frac{h}{2\pi} \cdot 2\pi}$$

$$= \frac{g \beta_N B_z}{h} \text{ Hz}.$$

We know, $\frac{\Delta E}{h} = \left| \frac{g \beta_N B_z}{h} \right|$ Hz

So, the larmor frequency is just equal to the frequency separation betn energy level.

(d)

(8) Explain why electron-nucleous coupling constans are much greater than those for nucleous-nucleous interactions. Sketch and discuss the ESR spectrum of methyl radical $\cdot CH_3$.

Answer:

The reason for such results are:

(i) An electron can approach with a nucleous more closely than nucleous-nucleous of electron

(ii) The electron's magnetic dipole is 1000 times larger than that of a nucleous.

The magnitude of coupling constant is directly related to the electron density.

$A = RP$

For H, $P = 1$, and $A = 0.05\,T$.

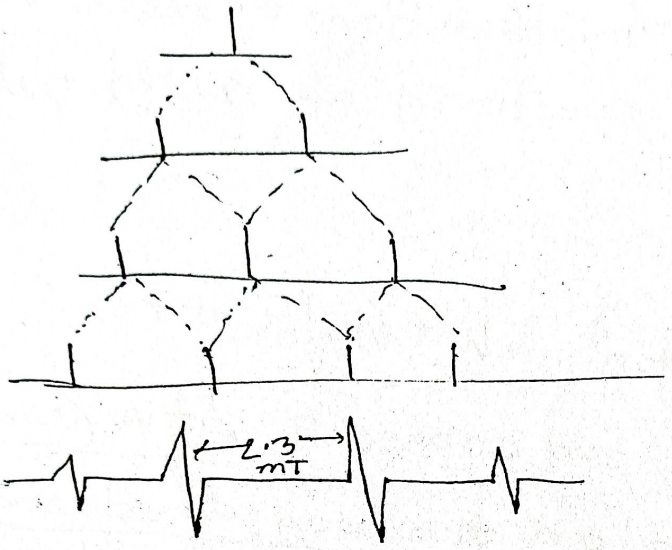
$R = 0.05\,T$.

$E = PR$

$2xNL+1$

E.S.R spectrum of $^{\bullet}CH_3$: $= 2 \times 3 \times \frac{1}{2} + 1$

1:3:3:1 quartet $= 4$



$$R = \frac{A}{R} = \frac{0.0023}{0.05}$$

≈ 0.046

$\approx 4.6\%$ For 1 H.

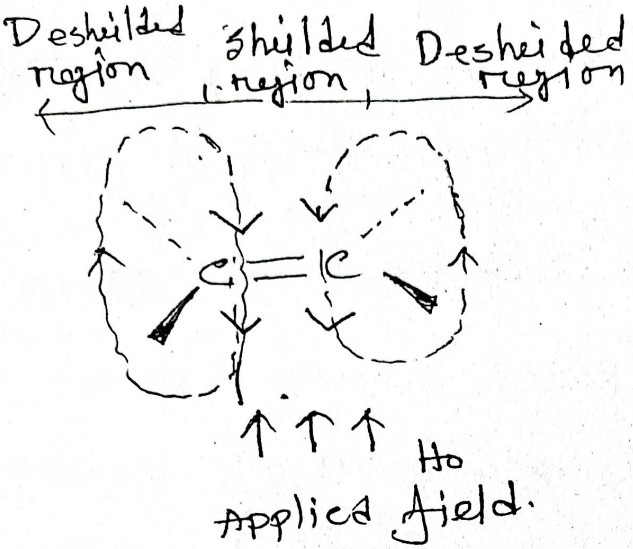
$\rho = 15\%$ For 3 H.

85% electron density force.

7.© Discuss the anisotropic effect on the chemical shift values.

Answer:

Anisotropic effect



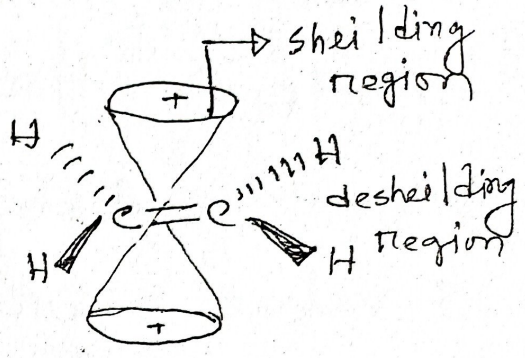


Fig:

Anisotropic effect is shown the compound which contain π electron. such as C=C, C≡O, aromatic compound with π electron and ~~appli~~ magnetic field is applied. for this reason, magnetic field is induced. when induced magnetic field is opposite to the applied magnetic field, then this region is more sheilded. If the induced magnetic field is aline to the applied magnetic field, and Desheided region is formed.

In this figure, cannonical form is also shown, the cannonical form is the sheilded region.

Consider, a example benzene contain π electron. Anisotropic effect also present and proton present in deshilded region and gives higher δ value = 7.27 ppm.

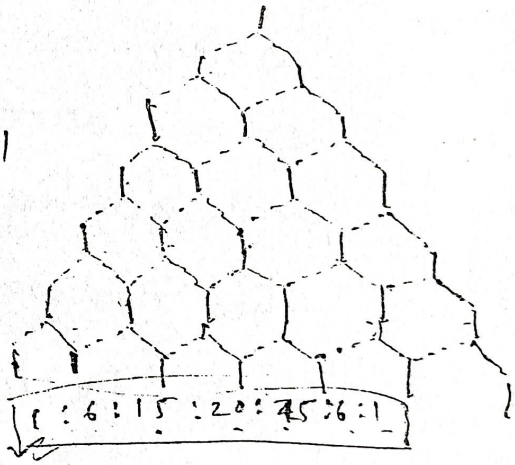
(f) Sketch the E.S.R spectrum of benzene radical. Find the total electron density for six H of this radical.

Answer:

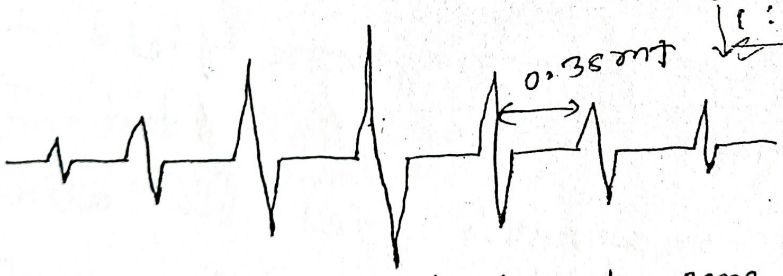
The no. of lines $2NI+1$

$= 2\times6\times\frac{1}{2}+1$

$= 7$



1:6:15:20:15:6:1



E.S.R spectrum of the benzene radical.

$A = \rho R \quad \rho = \frac{0.00038}{0.05} = 7.6\times10^{-3}$

For six H $= 6\times7.60\times10^{-3}$

$= 0.046 \approx 5\%$

For 6 e = 95%. For each e = 16%.

2015

7. (a) Give examples of nuclei with zero spin, integral spin and half-integral spin.

Answer:

All electrons have a spin of ½.

(i) Nuclei with both p and n even (hence charge and mass even) have zero spin.

Example: ^{4}He, ^{16}O, ^{12}C etc.

(ii) Nuclei with both p and n odd have integral spin.

Example spin = 1 — ^{2}H, ^{14}N

spin = 3 = ^{10}B.

(iii) Nuclei with odd mass have half integral spins

spin ½ = ^{1}H, ^{15}N

5/2 → ^{17}O

[2014]

(d) Explain why it is more convenient to refer ESR absorption in terms of its g factor. Why is the g factor value varied to a large extent in case of ionic crystals of transition metal ion?

Answer:

The spin energy levels of an electron are separated in an applied magnetic field, B_0, by an amount

$$\omega / \frac{\Delta E}{h} = \frac{g\beta B_0}{h} \text{ Hz.}$$

where, β is the Bohr magneton (9.273×10^{-24} JT^{-1}) and g is the Lande splitting factor or proportionality factor.

A resonance absorption will occur at a frequency $\nu = \frac{\Delta E}{h}$ Hz. The position of absorption varies directly with applied field and different e.s.r spectrometers operate at different fields, and this absorption also depent on g value.

Thus, rearranging the equation we have,

$$g = \frac{\Delta E}{\beta B_0} = \frac{h\nu}{\beta B_0}.$$

If resonance was observed at 0.34 MHz in a field of 0506.690T and g value of 2.0023.

For free radicals and some ionic crystals g values may be varies only some ± 0.003.

For a free electron - 2.00232

For ionic crystal of organic transition metal compound 1.4 - 3.0

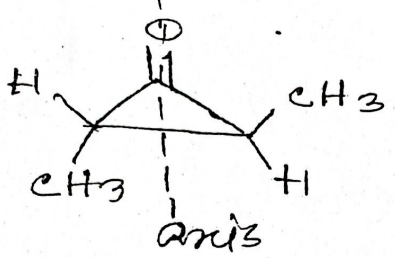
Large variation is occur due to spin-orbit coupling and zero-field spliting.

⑥ Differentiate betn chemically equivalent and magnetically equivalent nuclei with examples.

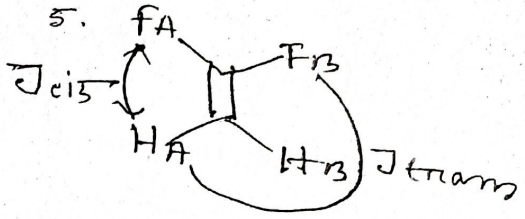
Answer:

Chemically equivalent	Magnetically equivalent
① If two or more nuclei equivalent by symmetry they are said to be chemically equivalent.	① Chemically equivalent nuclei which have the same chemical shift, do not split each other and give single NMR signal, are magnetically equivalent.
2. Only plane of symmetry or axis of symmetry renders nuclei chemically equivalent.	2. Only plane of symmetry can't determine that nuclei are magnetically equivalent or not.
3. chemically equivalent nuclei do not necessarily have identical chemical shift.	3. Magnetically equiv. nuclei must have identical chemical shift.
4. chemically equivalent nuclei may not have same coupling constant in all cases.	4. Magnetically equiv. nuclei must have equal coupling constant to all other nuclei in the molecule.

5. Chemically equiv.



Magnetically equiv.

5. 

CH_3 groups are chemically equivalent. Two $-CH_3$ carbon atoms yield single peak in ^{13}C NMR spectroscopy

5 cis and trans coupling being different in magnitude, there one not magnetically equivalent.

Tutorial question:

Define (i) coupling constant (ii) Hyperfine (iii) chemical shift.

Answer:

Coupling constant: The spacing in hertz (Hz) between the peaks is a constant is called coupling constant.

or, The magnitude of the separation is called coupling constant (J) and unit is Hz.

Coupling rules:

(i) equivalent nuclei do not interact.

(ii) coupling constant decreases with separation.

(iii) multiplicity given by the no. of attached equivalent protons (n+1)

(iv) multiple spin system → multiplicity (n+1) (n+1)

(v) coupling pattern and intensity follow Pascal's triangle

(vi) coupling constant are independent of applied field strength.

Hyperfine interaction:

ESR or EPR signal is split by neighbouring nuclei called hyperfine interaction.

Information get by heperfine

(i) coupling pattern same as NMR

(ii) The number of lines $2NI+1$

N = no. of equivalent nuclei

I = spin

(iii) Only determines no. of lines not the intensities

(iv) If only one nucleous interacting - all lines have equal intensity.

(v) If multiple nuclei interacting -

→ For spin ½, intensities follow binomial function-

⑤ Number and identity of nuclei

⑥ Distance from unpaired electron.

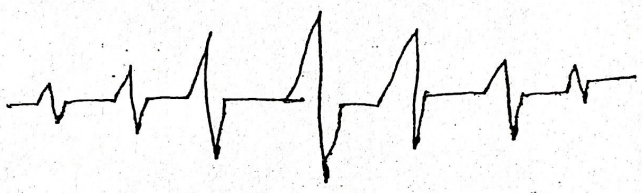
Example: For benzene radical—

C_6H_6. $I = \frac{1}{2}$

$N = 6$

No. of lines $= 2NI + 1$

$= 2 \cdot 6 \cdot \frac{1}{2} + 1 = 7$.



chemical shift: The value of δ is called the chemical shift.

$$\delta = \frac{\omega - \omega_{ref}}{\omega_{ref}} \text{ PPM}$$

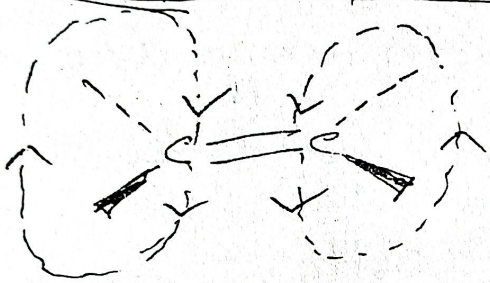
chemical shift value depends on some thing.

① Effect of magnetic anisotropy :-

In a π electron system, if external magnetic field is induced and for this reason π electron is circulation and induce

magnetic field and when induce magnetic field opposite to the oplied field, they are sheilded region and allign to the magnetic field in desheilded region.

Desheilded, sheiled, Desheilded



Induce magnetic field.

↑↑↑ (Applied field)

Consider a example : ethanol-

CH_3-CH_2-OH